$$\rho = \frac{1}{q(\mu_n \cdot n + \mu_p p)} = \frac{1}{q(\mu_n + \mu_p) n_i}$$

Si es intrínseco

$$n_i = c \cdot T^{3/2} e^{-\frac{E_g}{2KT}} \Rightarrow \rho = \frac{1}{c \cdot q(\mu_n + \mu_p)} (T)^{-3/2} e^{\frac{E_g}{2KT}} = c' (T)^{-3/2} e^{\frac{E_g}{KT}}$$

$$R(T) = \rho(T) \cdot \frac{\text{Largo}}{\text{Area}} = c''(T)^{-3/2} e^{\frac{E_g}{2KT}}$$

$$R_{298} = R(298) \qquad \frac{R(T)}{R_{298}} = \frac{\cancel{c''}(T)^{-3/2} \; e^{\frac{E_g}{2KT}}}{\cancel{c''}\,(298)^{-3/2} \; e^{\frac{E_g}{2K \cdot 298}}} =$$

$$R(T) = R_{298} \left(\frac{T}{298}\right)^{-3/2} \exp\left\{ \frac{E_g}{2K} \left( \frac{1}{T} - \frac{1}{298} \right) \right\}$$

$$R(T) = R_{298} \left(\frac{T}{298}\right)^{-3/2} \exp\left\{ -\frac{E_g}{2K} \left( \frac{1}{298} - \frac{1}{T} \right) \right\}$$

$$\left[ \text{Si} \quad \frac{T}{298} \sim 1 \right] \Rightarrow R(T) = R_{298} \exp\left\{ -\underbrace{\frac{E_g}{2K}}_{\beta} \left( \frac{1}{298} - \frac{1}{T} \right) \right\}$$

Elijo creer

$$\ln(R(T)) = \ln R_{298} - \beta \left( \frac{1}{298} - \frac{1}{T} \right) = a + \beta \cdot x$$

→ Ajustar la recta por python

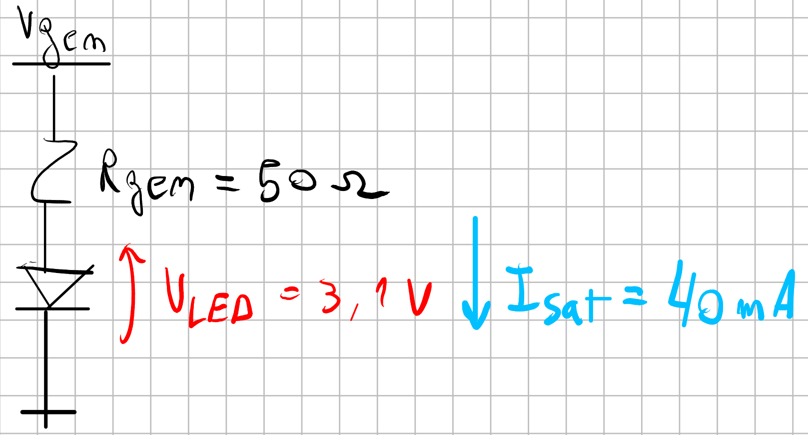
$$\beta \left( \frac{1}{298} - \frac{1}{T} \right) = \ln \frac{R_{298}}{R(T)} \Rightarrow \beta = \left( \frac{1}{298} - \frac{1}{T} \right)^{-1} \ln \frac{R_{298}}{R(T)}$$

$R_{238}$ lo sacamos de la simulación

$$R_{238} = \frac{553\ mV}{55,9\ \mu A} = \boxed{9892\ \Omega}$$

¿Posible SC? → del $\beta$ despejar Eq. $Eq = \beta . 2K$

---

1) 


$$V_{gen} = V_{LED(ON)} + R_{gen} \cdot I_{LED}$$

$V_{gen} = 5,1\ V$ (con el LED saturado)

Hacer los gráficos. ¿Se ve algo?

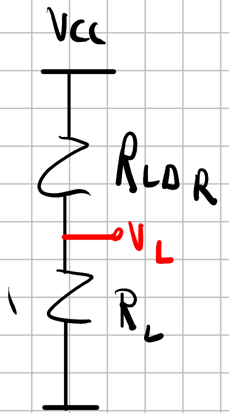
2) obtener $R_{LDR(osc)}$ (DDT)

3) $R_{LDR(osc)} = \rho \dfrac{L}{A} \Rightarrow \dfrac{L}{A} = \dfrac{R_{LDR(osc)}}{\rho}$ $\left(T = 300K \text{ según consigna}\right)$

$\rho = \dfrac{1}{q(\mu_n+\mu_p)n_i}$ $\quad n_i = cte\,(T)^{3/2}\, e^{-\frac{E_g}{2kT}}$ (¿o usar las masas efectivas?)

¿No está dopado? → El LDR es intrínseco, está en la consigna

4)




$R_{LDR(osc)} = \rho_{osc}\cdot\dfrac{L}{A} \to \rho_{osc} = \dfrac{A}{L}\cdot R_{LDR(osc)}$

$R_{LDR(ON)} = \rho_{luz}\cdot\dfrac{L}{A} \to \rho_{luz} = \dfrac{A}{L}\cdot R_{LDR(luz)}$

$V_L = \dfrac{R_L}{R_L+R_{LDR}}\cdot V_{cc} = \dfrac{1}{1+\frac{R_{DL}}{R_L}}\cdot V_{cc} \Rightarrow \dfrac{R_{LDR}}{R_L} = \dfrac{V_{cc}-V_L}{V_L}$

$\Rightarrow R_{LDR} = \dfrac{V_{cc}-V_L}{V_L}\cdot R_L$

$$\sigma(osc) = \rho_{osc}^{-1} = \frac{L}{A} \cdot \frac{1}{R_{LDR}(osc)}$$

$$\sigma(LUZ) = \rho_{LUZ}^{-1} = \frac{L}{A} \cdot \frac{1}{R_{LDR}(LUZ)}$$

$$\Delta\sigma = \sigma(LUZ) - \sigma(osc) = \frac{L}{A}\left(\frac{1}{R_{LDR}(LUZ)} - \frac{1}{R_{LDR}(osc)}\right)$$

$$R_{LDR}(LUZ) = \frac{V_{cc} - V_L(LUZ)}{V_L(LUZ)} \cdot R_L \; ; \; R_{LDR}(osc) = \frac{V_{cc} - V_L(osc)}{V_L(osc)} \cdot R_L$$

$$\boxed{\Rightarrow \Delta\sigma = \frac{L}{A}\left[\frac{V_L(LUZ)}{V_{cc} - V_L(LUZ)} \frac{1}{R_L} - \frac{1}{R_{LDR}(osc)}\right]}$$

5) Buscar en el Libro/Clase

6) $\delta n(t) = cte\, e^{-t/\tau}$

Para las distintas intensidades tiene que dar lo mismo, no depende de $\delta n$.

7) ¿ $g$ ? → Ecuación ambipolar.

$$\frac{d\,n(t)}{dt} = \alpha_r\left[n_i^2 - n(t).p(t)\right]$$

$$n(t) = n_0 + \delta n(t) = n_i + \delta n(t)$$

$$p(t) = n_i + \delta n(t)$$

$$\frac{d\,\delta n(t)}{dt} = \alpha_r\left[\cancel{n_i^2} - \cancel{n_i^2} - 2n_i\,\delta n(t) - \delta n(t)^2\right] = -\alpha_r\left[2n_i - \delta \cancel{n}(t)\right].\delta n(t)$$

Baja inyección: $n_i >> \delta n(t) \Rightarrow \frac{d\,\delta n(t)}{dt} = -\alpha_r\, 2n_i\, \delta n(t)$

$$\boxed{\frac{d\,\delta n(t)}{dt} = -\alpha_r\, 2n_i\, \delta n(t)} \Rightarrow \delta n(t) = \delta n(0)\, e^{-2\alpha_r n_i . t} = \delta n(0)\, e^{-t/\tau_n}$$

com $\tau_n = \frac{1}{2\alpha_r . n_i}$

$$\frac{dn}{dt} = -\frac{dF_n}{dx} + g_n - R_n = -\frac{dF_n}{dx} + g_n - \frac{n}{\tau_n}$$

$$J_n = q\,\mu_n\, n\,\mathcal{E} + q\,D_n \frac{\partial n}{\partial x} = q\,\mu_n . n\,\mathcal{E} + q\,V_{Th}\,\mu_n \frac{\partial n}{\partial x} \qquad \frac{D_n}{\mu_n} = V_{Th}$$

$$\frac{J_n}{(-q)} = -\mu_n . n . \mathcal{E} - V_{Th}\,\mu_n\, \partial n/\partial x = F_n \rightarrow \partial F_n/\partial x = -\mu_n \frac{\partial (n\mathcal{E})}{\partial x} - V_{Th}\,\mu_n\, \partial^2 n/\partial x^2$$

$$J_p/q = \mu_p . p\,\mathcal{E} - V_{Th}\,\mu_p\, \partial p/\partial x = F_p$$

$$\Rightarrow \frac{\partial n}{\partial t} = V_{Th}\mu_n \frac{\partial^2 n}{\partial x} + \mu_n \frac{\partial (n\mathcal{E})}{\partial x} + (g_n - R_n)$$

$$\Rightarrow \frac{\partial n}{\partial t} = D_n \frac{\partial^2 n}{\partial x} + \mu_n \left( \mathcal{E} \cdot \frac{\partial n}{\partial x} + n \frac{\partial \mathcal{E}}{\partial x} \right) + (g_n - R_n) \rightarrow \text{Ec de dif}$$

vale siempre

Condición de neutralidad (CNC) : $\delta n = \delta p \quad \forall x$ (ver. Neamen, cap. 6 p. 17 del pdf, p. 202 del libro)

$$\boxed{D' \frac{\partial^2(\delta n)}{\partial x^2} + \mu' \mathrm{E} \frac{\partial(\delta n)}{\partial x} + g - R = \frac{\partial(\delta n)}{\partial t}}$$

$$\boxed{\mu' = \frac{\mu_n \mu_p (p - n)}{\mu_n n + \mu_p p}} \qquad \boxed{D' = \frac{D_n D_p (n + p)}{D_n n + D_p p}}$$

Es importante que la temperatura sea cte para que ni no varíe con la temperatura

Ahora: SC intrínseco $\rightarrow$ $n = n_i + \delta n(t)$ $\quad p = n_i + \delta n(t)$ $\quad n = p$

$$\Rightarrow \mu' = 0 \quad , \quad D' = \frac{D_n D_p 2\cancel{n}}{(D_p + D_n)\cancel{n}} = 2 \cdot \frac{D_n \cdot D_p}{D_p + D_n}$$

$$\boxed{D' \cdot \frac{\partial^2 \delta n}{\partial x^2} + (g - R) = \frac{\partial \delta n}{\partial t}}, \quad \boxed{\text{con } D' = 2\,\frac{D_n D_p}{D_n + D_p}}$$

## Tiempo de vida

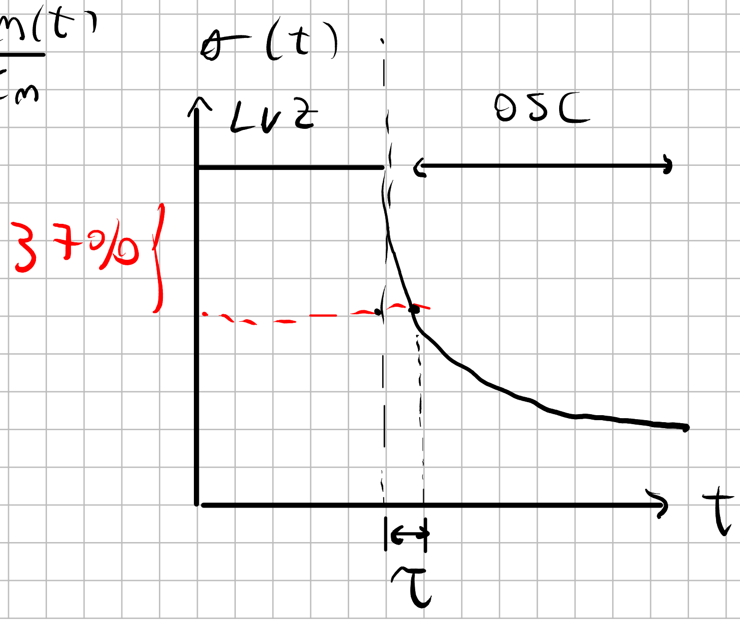
Si la iluminación es uniforme, no hay variaciones espaciales

$g(t) = A \cdot \delta(t) \rightarrow$ impulso en $t = 0$ s

Para $t > 0$ s: $\frac{\partial \delta n}{\partial t} = -R = -\frac{\delta n(t)}{\tau_n}$

$\Rightarrow \delta n(t) = \beta\, e^{-t/\tau} \propto \sigma$

$\sigma = q(\mu_n + \mu_p)(n + \delta n(t))$



"Corroborar la no dependecia de tau con la concentración de portadores en exceso"
--> Para todas las curvas tiene que dar igual

$$\frac{\partial \delta n}{\partial t} = g - \frac{\delta n(t)}{\tau}$$

En estado estacionario: $\partial \delta n / \partial t = 0$

$$g = \frac{\delta n_{est}}{\tau}$$

Similar a Neamen, ejemplos 6.2 y 6.3

$$\sigma_{(est)} = q(\mu_n + \mu_p)(n_i + \delta n_{(est)})$$

→ Estado estacionario, para cuando el LED está prendido.

$$\delta n_{(est)} = \frac{\sigma_{(est)}}{\mu_n + \mu_p} - n_i = \delta p_{(est)}$$